Universität Zürich[UZH]

ETH

Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

DaST

Data • (Systems+Theory)

VLG Computer Vision and Learning Group

Bachelor's Thesis

Submission of the thesis
June 8, 2026

Advancing (Sparse) Multi-View Depth Estimation

Fabio Di Meo
University of Zurich

Student-ID: 22-929-798
fabio.dimeo@uzh.ch

Supervisors:
Prof. Dr. Dan Olteanu,
Frano Rajič

Abstract

Zusammenfassung

Acknowledgements

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Contents

1

Introduction

Recovering the three-dimensional structure of a dynamic scene from visual observations is a foundational challenge in computer vision with direct implications for robotics and autonomous navigation [3]. Recent spatial foundation models—such as MASt3R-SfM [5], VGGT [8], PI3, and PI3X [10]—have achieved remarkable per-frame geometric fidelity by regressing dense 3D pointmaps directly from images. Yet in 4D settings, where temporal coherence across a video sequence is as critical as spatial accuracy, these models fail: assembled frame-by-frame, their outputs exhibit severe high-frequency jitter that renders them unsuitable for robotics applications requiring smooth, metric-consistent depth. Geometry-refinement approaches such as GGPT [2] attempt to correct these inconsistencies by grounding spatial predictions with sparse SfM guidance, but fail to generalise to sparse-view dynamic scenes, worsening Chamfer distance by 3–10×. Conversely, dedicated 4D architectures—VGGT4D [4] and MonoFusion [11]—sacrifice geometric precision to buy temporal smoothness, inflating Chamfer distance by 20–160 % and producing systematic alignment failures in sparse-view settings.

The central finding of this thesis is that a *decoupled* paradigm outperforms both families: take a strong per-frame 3D estimator, align its outputs into a common coordinate frame through a lightweight temporal alignment strategy, and apply a confidence-weighted, mask-aware Gaussian post-processing filter (Opt) to enforce static-scene consistency. This approach reduces jitter by 25–81 % across four models (MASt3R-SfM, VGGT, PI3, PI3X) while preserving geometry to within ±2–8 % Chamfer distance—a strictly superior trade-off to every native 4D architecture and geometry-refinement baseline we evaluate. At the two-camera extreme, VGGT combined with Opt beats the native 4D architecture VGGT4D on every single metric.

1.1 Background and Motivation

Classical multi-view reconstruction pipelines such as Structure-from-Motion (SfM) [7] produce accurate, metric-scale geometry in well-constrained settings: they rely on repeatable keypoint matching and bundle adjustment, which deliver precise camera poses and sparse point clouds when images share sufficient texture, overlap, and baseline. In sparse-view configurations of 2–4 cameras, however, these conditions frequently fail. Low inter-camera overlap means that few keypoints are visible in more than one view simultaneously [3]; short or degenerate baselines reduce parallax and destabilise triangulation [7]; and textureless surfaces—common on human skin, clothing, and manipulation objects—yield too few reliable correspondences for bundle adjustment to converge [9, 5]. The result is either outright reconstruction failure or incomplete, drift-prone geometry that cannot support downstream tasks such as grasping or object tracking.

Learning-based spatial models such as DUSt3R [9], PI3, and PI3X [10] overcome many of these failure modes by directly regressing dense 3D pointmaps from image pairs, using learned priors to fill in low-texture and low-overlap regions where classical methods collapse. These models achieve strong per-frame accuracy—PI3 and PI3X in particular deliver the highest accuracy and completeness at a strict 1 cm robotics threshold among all models we evaluate. Although they address many of the correspondence and reconstruction failures that limit classical SfM, they do not eliminate the broader challenge of producing reconstructions that are sufficiently stable for downstream robotic use. As purely spatial estimators, they process each frame independently and therefore provide no explicit mechanism for enforcing temporal consistency across a sequence. The resulting frame-to-frame instability can manifest as high-frequency jitter, scale fluctuations, or pose inconsistencies, with the severity varying substantially across architectures. Such instability can render otherwise accurate reconstructions unsuitable for downstream tasks including trajectory estimation, object tracking, and robotic manipulation.

GGPT [2] augments feed-forward reconstructions using geometry-guided refinement informed by sparse SfM-derived correspondences. While the method improves reconstruction quality on the static benchmarks for which it was designed, its behaviour under the sparse-view dynamic conditions considered in this thesis differs substantially. Across all evaluated backbone models, GGPT consistently reduces the measured static-scene jitter metric, often by an order of magnitude. However, this reduction is accompanied by large increases in Chamfer distance and inconsistent changes in accuracy and completeness. Since the degradation is already evident in the per-frame evaluation prior to temporal alignment, the observed jitter suppression is unlikely to arise from improved geometric fidelity. Instead, the results suggest that GGPT acts as a strong geometric regulariser in this setting, producing more temporally stable but less geometrically accurate reconstructions.

The natural response is to incorporate temporal modelling directly. VGGT4D [4] extends spatial transformers with motion cues, and MonoFusion [11] fuses monocular depth priors across time. Both enforce sequence-level consistency, but at a cost we measure precisely: VGGT4D inflates Chamfer distance by 20–160 % relative to its per-frame baseline, and MonoFusion suffers systematic alignment convergence failures in sparse-view settings, producing duplicated subject artefacts rather than coherent reconstructions.

None of these approaches simultaneously achieves geometric fidelity and temporal consistency under sparse-view dynamic conditions. These results motivate the central design principle of this work: temporal consistency and geometric fidelity are not in fundamental conflict—they simply require a targeted rather than global smoothing strategy.

1.2 Problem Statement and Contributions

We address the problem of temporally consistent, geometrically precise 4D reconstruction from sparse multi-view video (2–4 cameras) of dynamic scenes, evaluated on the DEX-YCB [1] and Hi4D [12] datasets. The main contributions of this work are:

- **Unified 4D Benchmarking Framework.** A structured evaluation pipeline that subjects six base models—MASt3R-SfM, VGGT, PI3, PI3X, VGGT4D, and MonoFusion—to identical data-loading, semantic masking, and metric-computation conditions across DEX-YCB [1] and Hi4D [12]. The framework measures both 3D per-frame and 4D sequence-level performance via Chamfer distance, accuracy and completeness at a 1 cm threshold, temporal jitter, drift, and camera pose error, with static/dynamic decomposition where applicable (see Chapter 3). It additionally evaluates the effect of GGPT [2] geometry refinement applied on top of all base models except MonoFusion.

- **Comparative Temporal Alignment Study.** An analysis of three strategies for lifting per-frame 3D reconstructions into a globally consistent 4D representation (see Chapter 3). The Reference strategy aligns every frame to a common anchor using static-scene correspondences and the Umeyama similarity transform. The Hierarchical strategy builds a binary tree of pairwise alignments, propagating consistency bottom-up. The Pose-Graph Optimisation strategy computes all $\mathcal{O}(T^2)$ pairwise edges, weights them by alignment reliability, and refines all frame poses simultaneously via robust geodesic averaging on SO(3). The Hierarchical strategy consistently delivers the best static-scene completeness and geometric stability; Pose-Graph Optimisation minimises the temporal consistency cost $\Delta_{\text{consistency}}$ (Equation 1.1) and best preserves dynamic-region accuracy.

$$\Delta_{\text{consistency}} = \text{Chamfer}_{\text{4D}} - \text{Chamfer}_{\text{3D}} \tag{1.1}$$

This quantity isolates the spatial degradation introduced purely by coordinate alignment, enabling direct comparison of alignment strategies and model families.

- **Confidence-Weighted Temporal Smoothing (Opt).** A ground-truth-free post-processing method that reduces frame-to-frame instability for any per-frame estimator without retraining (see **??**). Opt applies a Gaussian temporal filter to already-aligned pointmaps, weighted by model confidence and restricted to pixels classified as static by SAM2 [6]-derived masks—leaving dynamic pixels untouched to avoid suppressing real motion. A Pareto-frontier sweep over temporal bandwidth $\sigma \in \{0.0, 0.5, 1, 2, 4, 6, 8, 12, 20, 30\}$ and blending strength $\alpha \in \{0.0, 0.2, 0.5, 0.8, 1.0\}$ demonstrates that the optimal configuration is model-dependent: MASt3R-SfM benefits from a wider temporal window (σ=6, α=1.0), PI3 and PI3X from a moderate window (σ=4, α=1.0), and VGGT from a narrower window (σ=2, α=1.0), reflecting distinct jitter profiles across architectures. These per-model configurations achieve a 25–81 % jitter reduction while preserving geometry to within ±2–8 % Chamfer distance—a strictly superior trade-off to every native 4D architecture and geometry-refinement baseline evaluated in this thesis.

- **Empirical Characterisation of Architectural Families.** A systematic comparison of spatial-first models (PI3, PI3X, MASt3R-SfM, VGGT), the geometry-refinement approach GGPT, and native 4D architectures (VGGT4D, MonoFusion) reveals distinct and quantifiable failure modes under sparse-view dynamic conditions. PI3 and PI3X consistently achieve the strongest geometric performance across both DEX-YCB and Hi4D. MASt3R-SfM exhibits the greatest frame-to-frame instability, making it the model that benefits most from temporal refinement. VGGT provides a competitive balance between accuracy and stability. GGPT worsens Chamfer distance by 3–10$\times$ and produces inconsistent accuracy and completeness behaviour across architectures and view counts, while suppressing jitter to near-zero—a combination that makes it unsuitable for geometry-critical applications despite its apparent temporal stability. Native 4D architectures reduce temporal noise but consistently sacrifice geometric accuracy relative to spatial-first models. Together these findings establish the decoupled framework—strong spatial estimation, robust temporal alignment, and confidence-weighted temporal smoothing—as the most effective approach for sparse-view 4D reconstruction.

2

Literature Review and Background

3

Methodology: The Evaluation Framework

4

System Implementation

5 Results

6

Refinement Strategy

7

Discussion

8

Conclusions

Bibliography

[1] Yu-Wei Chao *et al.* "DexYCB: A Benchmark for Capturing Hand Grasping of Objects". In: *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2021.

[2] Yutong Chen *et al.* "GGPT: Geometry-Grounded Point Transformer". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026.

[3] R. I. Hartley and A. Zisserman. *Multiple View Geometry in Computer Vision*. Second. Cambridge University Press, ISBN: 0521540518, 2004.

[4] Yu Hu *et al.* *VGGT4D: Mining Motion Cues in Visual Geometry Transformers for 4D Scene Reconstruction*. 2025. arXiv: `2511.19971 [cs.CV]`. URL: `https://arxiv.org/abs/2511.19971`.

[5] Vincent Leroy, Yohann Cabon, and Jerome Revaud. *Grounding Image Matching in 3D with MASt3R*. 2024. arXiv: `2406.09756 [cs.CV]`.

[6] Nikhila Ravi *et al.* "SAM 2: Segment Anything in Images and Videos". In: *arXiv preprint arXiv:2408.00714* (2024). URL: `https://arxiv.org/abs/2408.00714`.

[7] Johannes Lutz Schönberger and Jan-Michael Frahm. "Structure-from-Motion Revisited". In: *Conference on Computer Vision and Pattern Recognition (CVPR)*. 2016.

[8] Jianyuan Wang *et al.* *VGGT: Visual Geometry Grounded Transformer*. 2025. arXiv: `2503.11651 [cs.CV]`. URL: `https://arxiv.org/abs/2503.11651`.

[9] Shuzhe Wang *et al.* "DUSt3R: Geometric 3D Vision Made Easy". In: *CVPR*. 2024.

[10] Zihan Wang *et al.* "MonoFusion: Sparse-View 4D Reconstruction via Monocular Fusion". In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. Oct. 2025, pp. 8252–8263.

[11] Zihan Wang *et al.* “MonoFusion: Sparse-View 4D Reconstruction via Monocular Fusion”. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. Oct. 2025, pp. 8252–8263.

[12] Yifei Yin *et al.* “Hi4D: 4D Instance Segmentation of Close Human Interaction”. In: *Computer Vision and Pattern Recognition (CVPR)*. 2023.

List of Acronyms

A

Artifacts Availability

The artifacts associated with this thesis can be found in the ifi-GitLab[1]. Within this repository, there are two main folders: one containing the code to fine-tune the model, and the other containing the code to generate the training data, including all the different datasets. The README file provided in the repository contains step-by-step instructions on how to set up the environment used in this thesis, install all necessary dependencies, and run the source code.

Upload your Code to GitLab rather than attaching it to the PDF.

[1] https://gitlab.ifi.uzh.ch/ddis/Students/Theses/<NAME>

Declaration of independence for written work

I hereby declare that I have composed this work independently and without the use of any aids other than those declared (including generative AI such as ChatGPT). I am aware that I take full responsibility for the scientific character of the submitted text myself, even if AI aids were used and declared (after written confirmation by the supervising professor). All passages taken verbatim or in sense from published or unpublished writings are identified as such. The work has not yet been submitted in the same or similar form or in excerpts as part of another examination.

Important: Update this text with the latest **Declaration of independence for written work** from `https://www.ifi.uzh.ch/en/studies/msc-info.html`. Edit me in `resources/declaration_en.tex`

Zürich, June 8, 2026

Signature of
Fabio Di Meo